GOVERNMENT OF KARNATAKA
Department of Collegiate and Technical Education
SRI JAYACHAMARAJENDRA GOVERNMENT POLYTECHNIC
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Bengaluru – 560001

2025–2026

A REPORT ON

"INTERNSHIP"

Submitted in partial fulfilment
of the requirements for the award of the

Diploma in Computer Science and Engineering

BY

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ACKNOWLEDGEMENT

We feel very happy to be a part of **Sri Jayachamarajendra Government Polytechnic,** Bangalore, and the **Department of Computer Science and Engineering**. This institution has given us the opportunity to acquire technical knowledge that has helped us immensely over the past three years. We sincerely thank our **Principal, Sri Harish**, for his unwavering support and for providing an excellent learning environment.

I am deeply grateful to our Head of the Department**, Mrs. Haritha S M**, for her insightful inputs and constructive feedback, which significantly contributed to my professional growth during this internship.

I would like to express my deepest appreciation to my industry supervisor **Mrs.Savitha Jawale** of **TestHeapStack Software Solutions Pvt Ltd**, whose expertise and guidance have been indispensable in providing direction to my project development efforts. Their patience, encouragement, and insightful feedback have helped me overcome challenges and achieve the desired results.

I extend my sincere thanks to my Cohort Owner **Sowmya C J**,Selection Grade Lecturer-I at SJ (Govt) Polytechnic College, for their constant monitoring and academic support throughout the internship period.

I would like to extend my heartfelt thanks to my **Parents** and **Well-Wishers** for their constant encouragement and unwavering support. Their love, belief, and sacrifices have been the driving force behind my success in this internship.

I am immensely grateful to my friends, whose constant support and collaborative spirit have made this journey truly enjoyable. Their willingness to share knowledge and offer constructive feedback has been invaluable in shaping the outcome of this work.

Sincerely,

Shrilakshmi A S

EXECUTIVE SUMMARY

This internship at **TestHeapStack Software Solution Pvt. Ltd., Bengaluru** was focused on the practical application of Artificial Intelligence and Machine Learning technologies to solve real-world problems in digital and healthcare domains. This report presents a comprehensive overview of the 16-week On Job Training experience, covering the organizational profile, technical skill development, project implementations, and key professional insights gained during the internship period.

The internship was structured around two major project deliverables. The first project involved the development of a **Fabricated News Prediction System**, an AI-based solution designed to identify whether a news article is *real* or *fake*. The system utilized Natural Language Processing (NLP) techniques such as text preprocessing, tokenization, and TF-IDF vectorization, along with machine learning algorithms like Logistic Regression for classification. The model was integrated into a Streamlit-based web application, enabling users to input news content and receive real-time predictions, thereby helping to combat misinformation.

The second project focused on building an AI-powered **Vitamin Deficiency Detection System**, aimed at identifying possible vitamin deficiencies based on user symptoms and inputs. The system leveraged machine learning techniques to analyze symptom patterns and provide preliminary insights into potential deficiencies such as Vitamin A, B, C, and D. A user-friendly interface was developed using Streamlit, allowing users to interact with the system and obtain quick health-related suggestions.

Throughout the internship, exposure to real-world development workflows, including dataset preparation, data preprocessing, model training, hyperparameter tuning, evaluation, and deployment, provided valuable practical experience. Working in a collaborative environment and following iterative development practices enhanced problem-solving skills and deepened the understanding of designing AI solutions tailored to user needs.

In conclusion, the internship at **TestHeapStack Software Solution Pvt. Ltd.** offered a valuable and transformative learning experience at the intersection of Natural Language Processing and healthcare-based AI applications. The technical skills and industry exposure gained during this period have established a strong foundation for pursuing a career in artificial intelligence and machine learning.

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ABBREVIATIONS

- **AI** – Artificial Intelligence
- **ML** – Machine Learning
- **NLP** – Natural Language Processing
- **TF-IDF** – Term Frequency – Inverse Document Frequency
- **NER** – Named Entity Recognition
- **XAI (Explainable AI)** – AI systems designed to provide transparent and interpretable results
- **ROC-AUC** – Receiver Operating Characteristic – Area Under Curve
- **LR** – Logistic Regression
- **NB** – Naive Bayes Algorithm
- **API** – Application Programming Interface
- **UI** – User Interface
- **HTML** – HyperText Markup Language
- **CSS** – Cascading Style Sheets
- **JS** – JavaScript
- **JSON** – JavaScript Object Notation
- **SQL** – Structured Query Language
- **CSV** – Comma-Separated Values
- **PDF** – Portable Document Format
- **DOCX** – Microsoft Word Document Format
- **IDE** – Integrated Development Environment
- **VS Code** – Visual Studio Code

- **IT** – Information Technology
- **OJT** – On Job Training
- **CRUD** – Create, Read, Update, Delete

CHAPTER - 1

COMPANY PROFILE

Figure 1.1: **TestHeapStack Software Solutions —** ***Company Logo***

1.1Overview of the Organization

Test Heap Stack Solutions was founded in the year 2023. Recognizing the burgeoning potential of the internet, the company started as a small web development firm, catering to local businesses in its home city. As the demand for web solutions grew, Test Heap Stack expanded its services to include e-commerce development, content management systems, Machine Learning, Data Learning, Artificial Intelligence, Quality Assurance, and custom software applications. The company quickly gained a reputation for delivering robust and scalable solutions.

1.2Vision And Mission

Fig 1.2 (Vision)

Vision statement: To become a trusted technology partner by delivering innovative, efficient, and intelligent software solutions that empower businesses and individuals.

Fig 1.2.1(Mission)

Mission Statement:

* To develop high-quality software and AI/ML solutions that solve real-world problem

* To continuously adopt emerging technologies and industry best practices.

* To ensure customer satisfaction through reliable products and strong technical support.

* ORGANIZATION STRUCTURE

Fig 1.3(Organization Structure)

Director & Key Decision-Maker

The company is led by Savita Prasad Jawale, who serves as the sole director. In an OPC, the director holds primary responsibility for strategic decisions, financial oversight, and overall management.

Core Technical Teams

- **Web Development & E-commerce Team:** Handles website creation, online store development, and CMS solutions.
- **Quality Assurance Team**: Ensures reliability, scalability, and performance of software products.
- **Custom Software Development Team**: Works on tailored solutions for clients across industries.

Support Functions

- Internship & Talent Development Programs: The company actively recruits freshers and interns, integrating them into project teams to build skills and support delivery.
- Client Relations & Business Development: Manages customer engagement, project requirements, and partnerships.
- Administrative & Finance: Oversees compliance, capital management, and operational efficiency.

ROLES AND RESPONSIBILITIES OF PERSONNEL

Fig 1.4(Role and Responsibilities)

1. Leadership

Director (Savita Prasad Jawale)

• Provides strategic vision and overall direction.
• Oversees financial management, compliance, and corporate governance.
• Makes final decisions on partnerships, expansion, and technology adoption.

2. Core Technical Teams

- Web Development & E-commerce Team

o Designs and develops websites, online stores, and CMS platforms.

o Ensures responsive, user-friendly, and secure digital experiences.

• AI/ML & Data Learning Team

o Builds machine learning models and AI-driven applications.

o Works on data analysis, predictive modeling, and automation solutions.

• Custom Software Development Team

o Creates tailored applications to meet client-specific needs.

o Focuses on scalability, performance, and integration with existing systems.

• Quality Assurance Team

o Conducts testing to ensure reliability, security, and efficiency.

o Implements automated testing frameworks and continuous improvement practices.

PRODUCTS AND MARKET PERFORMANCE

Fig 1.5 (Products)

Products:

- Custom Software Applications – Tailored solutions for businesses across industries.
- Web Applications & Websites – Responsive, scalable, and user-friendly platforms.

Services:

- Web Development – End-to-end design and development of websites and apps.
- Software Quality Assurance – Testing and validation to ensure reliability and performance.

Market Performances:

Fig 1.5.1 (Market Performances)

1. **Technical Training** :

- Designed to provide hands-on exposure to industry tools and frameworks.
- Helps participants gain practical knowledge aligned with current market needs.

2. **Certificate Courses**

- Structured learning modules with completion certificates recognized by industry peers.
- Focus areas include web technologies, programming languages, AI/ML applications, and QA practices.

CHAPTER 2

ON JOB TRAINING – 1:

FABRICATED NEWS PREDICTION SYSTEM

2.1 Role and Overview

During the first phase of my internship at **TestHeapStack Software Solutions Pvt. Ltd.**, I was assigned the role of **AI Engineer Intern**. My primary assignment was the complete design and development of a **Fabricated News Prediction System** — an AI-powered application that automatically classifies news articles as **Real or Fake** by analyzing textual content.

The system was designed to address the increasing spread of misinformation and fabricated news on digital platforms and social media. By applying **Machine Learning and Natural Language Processing (NLP)** techniques, the system analyzes linguistic patterns and textual features of news articles to predict whether the information is genuine or fabricated.

This project aligns with the organization's focus on building **intelligent AI-based solutions** capable of solving real-world problems through automation and data analysis.

Layer / Component	Technology Used	Purpose / Description
Programming Language	Python	Core development language used for implementing ML models, backend logic, and data processing
Frontend	Streamlit	Provides an interactive user interface for entering news text and displaying prediction results
Backend Framework	Flask	Handles API requests, integrates ML model, and manages application logic
Machine Learning Models	Logistic Regression, Naïve Bayes	Used for classification of news into Real or Fake categories
NLP Techniques	TF-IDF Vectorization	Converts textual data into numerical features for model training and prediction
Data Processing Libraries	Pandas, NumPy	Used for data cleaning, preprocessing, and manipulation
Model Serialization	Joblib	Saves and loads trained ML models for efficient deployment
Dataset	Kaggle Fake News Dataset	Used for training and evaluating the model
Visualization (Optional)	Matplotlib, Seaborn	Used for data visualization and analysis during development
Development Environment	Jupyter Notebook / VS Code	Used for coding, testing, and experimentation
Version Control (Optional)	Git & GitHub	Manages code versioning and collaboration

Table 2.1:Fabricated News Prediction System — Technology Stack Summary

2.2 Application of Skills and Technical Knowledge

During this internship phase, I applied several technical skills related to **Machine Learning, Natural Language Processing, and web application development**. Python was used as the primary programming language for implementing the entire system, including data preprocessing, model training, and user interface development.

Machine Learning and NLP Implementation

Natural Language Processing techniques were applied to preprocess and analyze the news dataset. The preprocessing stage involved **text cleaning, tokenization, stop-word removal, and normalization** to prepare the textual data for machine learning analysis.

TF-IDF (Term Frequency–Inverse Document Frequency) was used for feature extraction, converting text data into numerical vectors that could be processed by machine learning models. This technique helps identify important words in news articles and reduces the impact of commonly used but less meaningful words.

A machine learning classification model was trained using the processed dataset to categorize news articles into **Real or Fake categories**. The trained model learns patterns from previously labeled news data and predicts the authenticity of new news articles provided by users.

User Interface Development

Streamlit was used to build an interactive web interface that allows users to input or paste news articles into the system. Once the user submits the text, the system processes the input, applies the trained model, and displays the prediction result indicating whether the news is **Real or Fake**.

The interface was designed to be simple and user-friendly, enabling easy interaction with the prediction system.

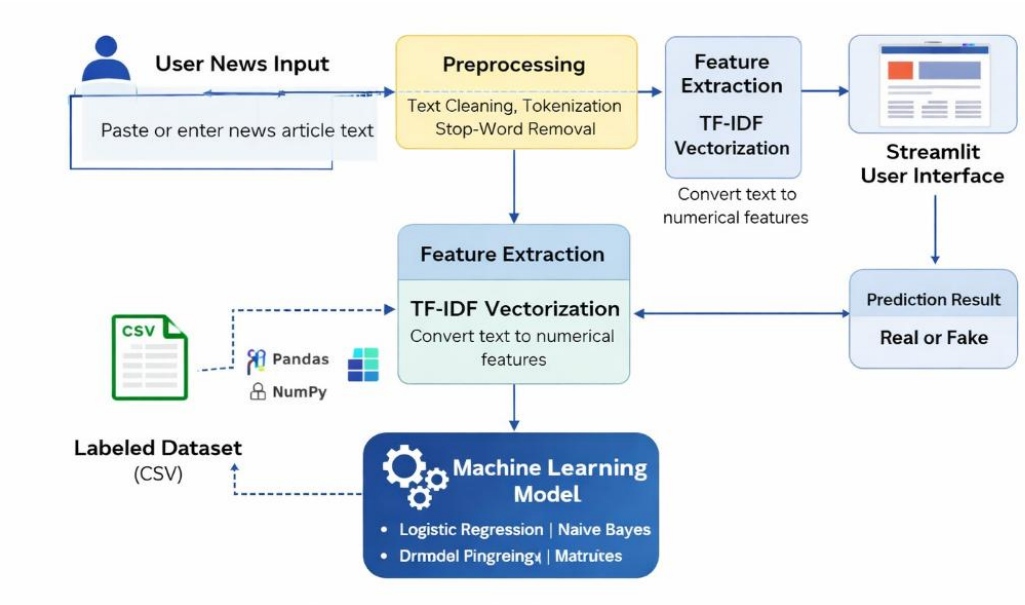

Figure 2.1: Fabricated News Prediction System Architecture

2.3 Performance on Assigned Tasks and Project

The project was implemented through a structured development process including dataset preparation, model development, and interface integration. The following components were successfully designed and implemented:

- Dataset collection and preprocessing for training the prediction model
- Text cleaning, tokenization, and stop-word removal using NLP techniques
- Feature extraction using TF-IDF vectorization
- Machine learning model training and evaluation
- News classification into Real or Fake categories
- Integration of prediction model with Streamlit interface
- User input processing and real-time prediction display
- Testing and debugging of the complete system

Feature	Description
Real-Time News Classification	Allows users to input news text and instantly receive predictions (Real or Fake) using trained ML models
User-Friendly Interface	Simple and interactive UI built with Streamlit for easy text input and result visualization
Text Preprocessing Pipeline	Performs cleaning, tokenization, stopword removal, and normalization of news content before analysis
TF-IDF Feature Extraction	Converts textual data into numerical vectors for accurate machine learning predictions
Multiple ML Models Support	Supports models like Logistic Regression and Naïve Bayes for better classification performance
Model Deployment	Trained model integrated with Flask backend for real-time prediction capability
High Accuracy Prediction	Achieves reliable classification results based on trained dataset and optimized model
Scalable Architecture	Modular design allows easy addition of advanced models (e.g., Deep Learning, BERT)
Fast Processing Speed	Optimized pipeline ensures quick response time for user queries
Dataset Handling	Efficient processing of large datasets using Pandas and NumPy

Table 2.2: Fabricated News Prediction System — Key Features Delivered

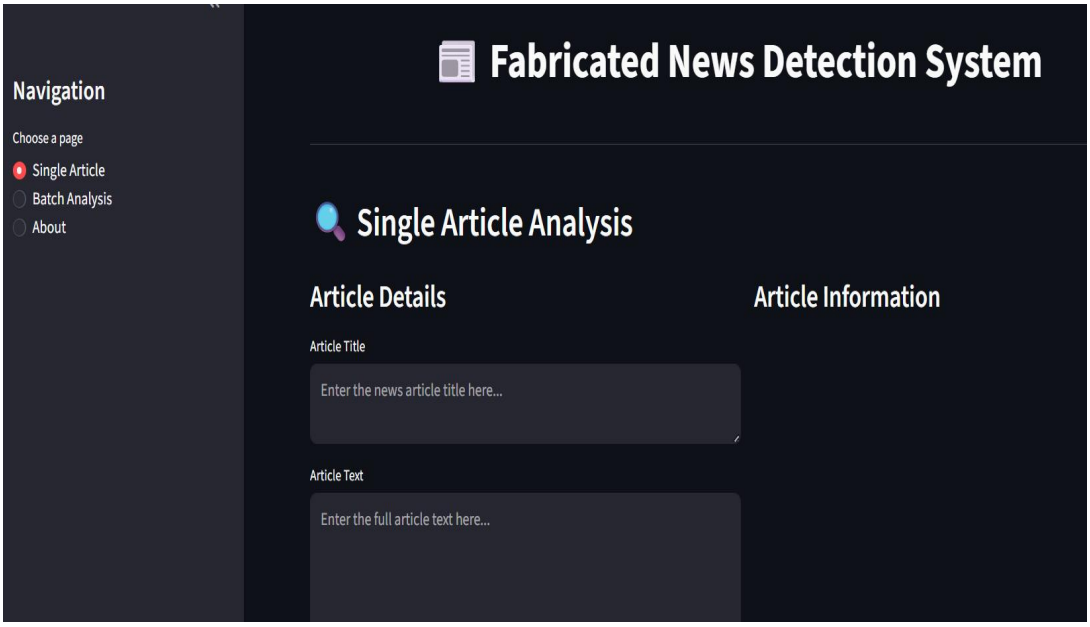

Figure 2.2: Fabricated News Prediction System — Streamlit UI: News Input and Prediction Result

2.4 Value Addition to the Organization

AI-Based News Verification Tool:

The developed system demonstrates how Artificial Intelligence can be used to automatically identify fabricated news, helping reduce the spread of misinformation across digital platforms.

Practical Application of Machine Learning:

The project showcases the practical use of machine learning techniques for solving real-world problems related to information verification and digital media analysis.

Reusable ML Model Architecture:

The modular architecture used in this project allows the system to be further extended with larger datasets, improved models, and integration with other web applications.

Improved Awareness of Misinformation:

The system highlights the importance of verifying online information and demonstrates how AI technology can assist users in identifying potentially misleading news articles.

CHAPTER 3

ON JOB TRAINING – 2:

VITAMIN DEFICIENCY DETECTION SYSTEM

3.1 Role and Overview

During the second phase of my internship at **Samai Infotech Solutions Pvt. Ltd.**, I was assigned the role of **AI Solution Developer Intern**. My primary assignment was the design, development, and integration of an **AI-Powered Vitamin Deficiency Detection System** — a smart web-based healthcare application that analyzes user symptoms and predicts possible vitamin deficiencies using Artificial Intelligence.

This system helps users understand potential health issues based on their symptoms and provides recommendations such as suggested foods, precautions, and health guidance. The application integrates AI-powered analysis with a user-friendly web interface to provide quick and accessible health insights.

This project demonstrates the organization's commitment to developing **intelligent AI-driven healthcare support applications** that simplify basic health analysis and improve digital health awareness.

Component	**Technology Used**
Primary Language	Python
Backend Framework	Flask
Frontend	HTML, CSS
AI Integration	DeepSeek API

Data Handling	JSON
Data Processing	Python
API Communication	REST API
Prediction System	AI-based symptom analysis
Deployment	Flask Web Application

3.2 Application of Skills and Technical Knowledge

During this phase of the internship, I applied several technical skills related to AI application development, backend programming, and API integration.

System Architecture

The Vitamin Deficiency Detection System was developed as a web-based application using Flask as the backend framework. The system allows users to enter their symptoms through a simple interface and receive AI-generated predictions related to possible vitamin deficiencies.

The backend processes user input, communicates with an AI-based API, and returns prediction results to the user.

Database Architecture

The system uses **JSON-based data storage** to manage prediction records and user interactions efficiently. Instead of a traditional database, structured JSON files were used to store and organize application data.The **Prediction Data** file stores information related to user symptom inputs and the prediction results generated by the AI system. It includes details such as symptoms entered by the user, predicted vitamin deficiency, severity level, recommended foods, precautions, and confidence score returned by the AI model.The **Feedback Data** file stores user feedback about the prediction results and system usability. This helps in analyzing user responses and improving the system performance in future

updates.The application also maintains **prediction history**, allowing the system to track previous user interactions and predictions.

Table 3.2: Vitamin Deficiency Detection System — AI Modules and Techniques Used

Module	Technique / Technology Used	Description
Image Input Module	Image Upload / Camera Input	Allows users to upload or capture images (skin, nails, eyes, etc.) for analysis
Image Preprocessing	OpenCV, Image Resizing, Normalization	Enhances image quality and prepares it for accurate model processing
Feature Extraction	Convolutional Neural Networks (CNN)	Extracts important visual features related to vitamin deficiency symptoms
Classification Model	Deep Learning (CNN / Transfer Learning)	Classifies images into different types of vitamin deficiencies (e.g., Vitamin A,B,C,D)
Dataset Handling	Image Datasets, Data Augmentation	Uses labeled datasets and augmentation techniques to improve model performance
Prediction Engine	TensorFlow / Keras	Runs trained models to generate deficiency predictions
Result Interpretation	Rule-Based Logic / Label Mapping	Converts model output into understandable deficiency results
User Interface	Streamlit / Web UI	Displays predictions and allows user interaction
Backend Integration	Flask API	Connects frontend with AI model for real-time processing
Visualization (Optional)	Matplotlib	Displays graphs or highlighted regions for better understanding

API Integration

The system integrates an AI-based API that performs intelligent analysis of symptoms. The Flask backend sends symptom data to the API and receives prediction results.

The API communication uses REST requests, where the backend sends symptom data and receives structured results.

Error Handling and System Reliability

To improve system reliability, a fallback mechanism was implemented. If the AI API fails or encounters an error, the system activates a backup logic to ensure the application continues to function.

This improves system stability and prevents interruption in user experience.

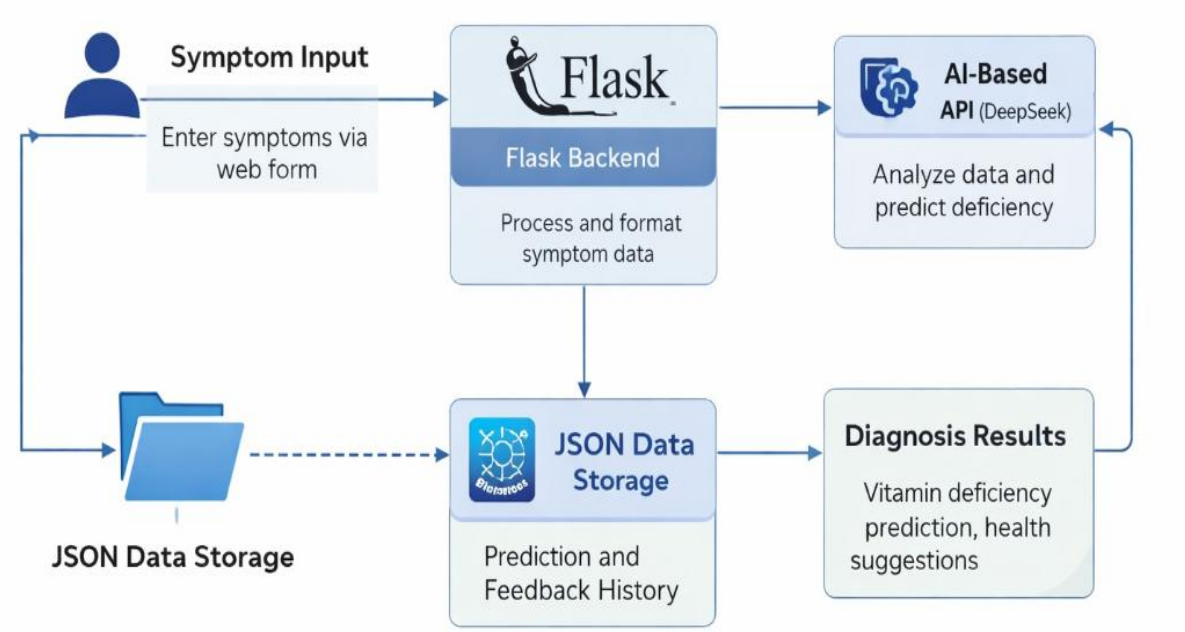

Figure 3.1: Vitamin Deficiency Detection System Architecture

3.3 Performance on Assigned Tasks and Project

The **Vitamin Deficiency Detection System** was successfully developed and tested with all planned features implemented. The project was completed through a structured

development approach including system design, backend development, AI integration, and testing.

Key features implemented include:

• User-friendly web interface for symptom input
• Backend development using Flask framework
• AI-based symptom analysis using API integration
• JSON-based response handling and data processing
• Prediction result display with health suggestions
• Storage of prediction history using JSON files
• Chatbot-based user guidance for symptom entry
• Error handling and API fallback mechanism
• Testing with different symptom inputs
• System deployment as a Flask web application

Feature	Description
Image-Based Detection	Allows users to upload images (skin, eyes, nails, etc.) to detect possible vitamin deficiencies
Automated Analysis	Uses AI models to automatically analyze visual symptoms without manual intervention
Real-Time Prediction	Provides instant results after image processing and model inference
Multiple Deficiency Classification	Detects various deficiencies such as Vitamin A, B, C, and D
User-Friendly Interface	Simple and interactive UI for easy image upload and result display
High Accuracy Detection	Utilizes deep learning models (CNN/Transfer Learning) for reliable predictions
Preprocessing Pipeline	Enhances image quality through resizing, normalization, and filtering techniques

Table 3.3: Vitamin Deficiency Detection System — Key Features Delivered

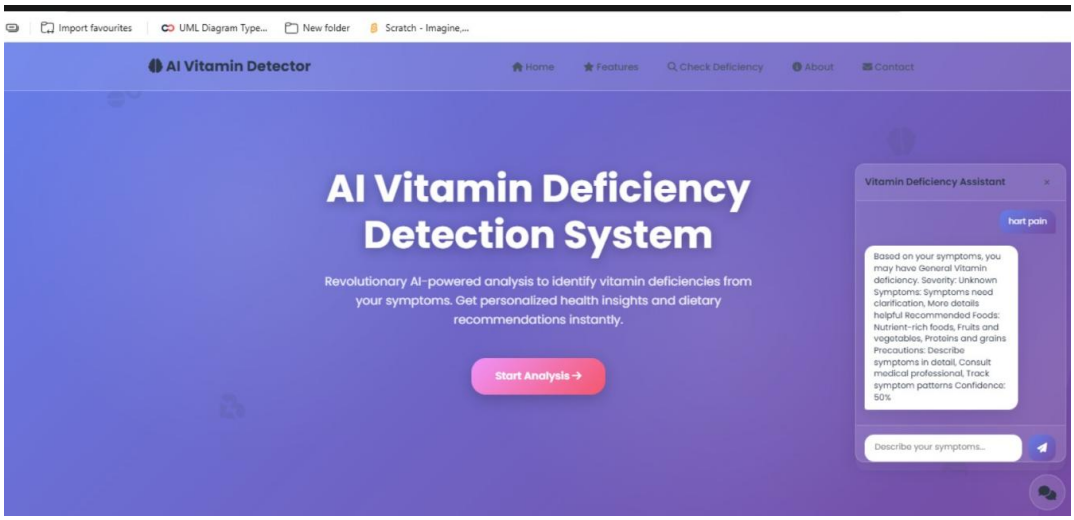

Figure 3.1 Architecture diagram for Vitamin Deficiency Detection System

3.4 Value Addition to the Organization

The Vitamin Deficiency Detection System provides significant value to the organization by demonstrating the practical use of Artificial Intelligence in healthcare support applications.

Improved Health Awareness

The system allows users to understand possible vitamin deficiencies based on their symptoms. This improves health awareness and encourages preventive care by providing instant suggestions.

AI-Based Intelligent Analysis

By integrating AI-based symptom analysis, the application can quickly evaluate health conditions and generate structured recommendations. This reduces manual effort and enables faster health guidance.

User-Friendly Healthcare Support Tool

The application provides a simple and accessible interface where users can enter symptoms and receive results without requiring medical expertise. The chatbot interaction further improves usability and makes the system interactive.

Demonstration of AI Application Development

The project demonstrates how AI technologies, API integration, and web development can be combined to create intelligent healthcare applications. This highlights the organization's capability to build innovative AI-based solutions for real-world problems.

Overall, the project supports the development of AI-driven digital healthcare tools and showcases the potential of intelligent systems in improving health awareness and accessibility.

CHAPTER 4 – USE CASES

4.1 Use Case 1 — Fabricated News Prediction System

4.1.1 Description

The **Fabricated News Prediction System** is designed to analyze news articles or headlines and determine whether the news is **Real or Fake** using Machine Learning techniques. The system accepts news text as input, processes the content using Natural Language Processing (NLP), and predicts the authenticity of the news using a trained machine learning model.

The system helps users quickly identify misleading or fabricated information circulating on digital platforms. By automating news verification, the application supports responsible information consumption and reduces the spread of misinformation.

4.1.2 Use Case Description

Actor	Role
User (Student / Reader / Researcher)	Enters news article or headline to check its authenticity
Fake News Prediction System	Processes the input news text and manages the overall prediction workflow
Machine Learning Model	Analyzes extracted text features and classifies news as Real or Fake
Dataset (Training Data)	Provides labeled data used for training and evaluating the ML model
Streamlit Web Interface	Enables user interaction through a simple and user-friendly web interface

4.1.3 Inputs

•News article or headline enterd by the user

•Pre-trained machine learning model

• News dataset used for model training

4.1.4 Process

1. User enters a news article or headline through the Streamlit web interface.
2. The system receives the input text and performs preprocessing.
3. Text preprocessing includes converting text to lowercase, removing punctuation, and cleaning unwanted characters.
4. The cleaned text is converted into numerical features using **TF-IDF Vectorization**.
5. The generated feature vector is passed to the trained machine learning model.
6. The machine learning model analyzes the text patterns and linguistic features.
7. The model predicts whether the news article is **Real** or **Fake**.
8. The prediction result is returned to the system.
9. The result is displayed to the user through the web interface.

10. The user can analyze the prediction and verify the credibility of the news article.

11. The user can analyze the prediction and verify the credibility of the news article.

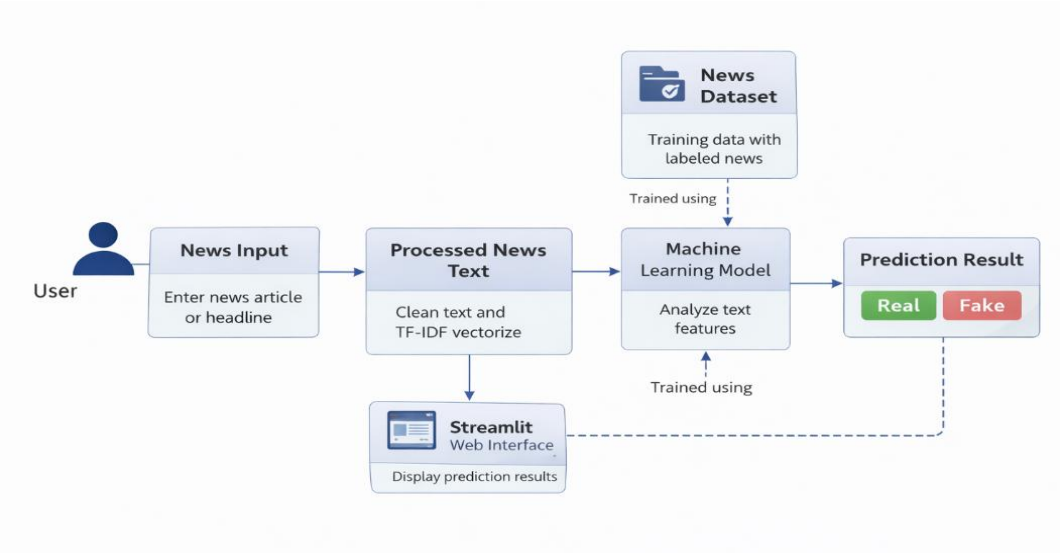

Figure 4.1: Fabricated News Prediction System — Workflow Diagram

4.1.5 Real-World Example

A user reads a suspicious news headline on social media and wants to verify whether it is real or fake. The user copies the headline and enters it into the system's input box. The system processes the text, extracts important features using TF-IDF vectorization, and sends the features to the trained machine learning model.

The model analyzes the text patterns and predicts whether the news is **Real** or **Fake**. The result is displayed instantly, helping the user identify whether the news is trustworthy.

4.1.6 Output

- Prediction result showing whether the news is Real or Fake
- Processed news analysis result
- User friendly prediction display on the web interface

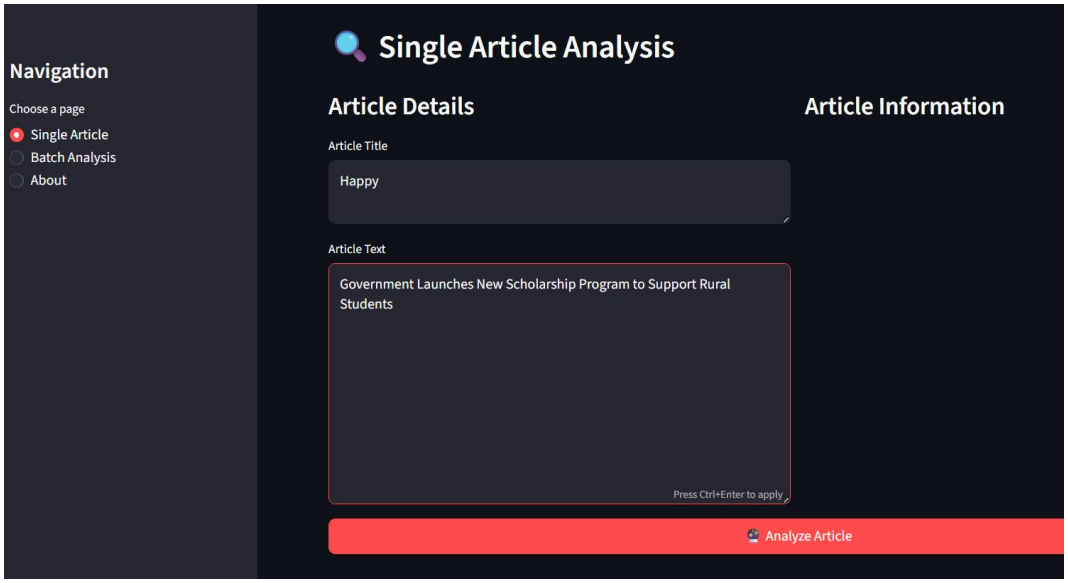

Figure 4.2: Fabricated News Prediction System — News Input and Prediction Result

4.1.7 Advantages

- Helps detect and reduce the spread of fake news on digital platforms
- Provides quick verification of news authenticity
- Uses machine learning techniques for intelligent text analysis
- User-friendly web interface for easy interaction
- Supports responsible information consumption

4.1.8 Limitations

- Prediction accuracy depends on the quality and size of the training dataset
- The system may misclassify very short or ambiguous news headlines
- Only text-based analysis is performed; images and videos are not analyzed
- Requires periodic dataset updates to adapt to new fake news patterns

4.1.9 Future Enhancements

- Integration of deep learning models for improved prediction accuracy
- Support for multilingual news analysis

- Detection of fake news from social media posts and websites
- Integration of image and video analysis for multimedia news verification
- Development of a browser extension for real-time fake news detection

4.1.10 Conclusion

During my internship, I successfully developed the Fabricated News Prediction System, a machine learning-based application designed to detect fake news articles automatically. The system demonstrates how Natural Language Processing and Machine Learning techniques can be applied to solve real-world problems related to misinformation.

This project provided practical experience in text preprocessing, feature extraction, machine learning model training, and web application development using Streamlit. The developed system highlights the potential of AI-driven tools in promoting trustworthy information and reducing the spread of misleading news.

4.2 Use Case 2 — Vitamin Deficiency Detection System

4.2.1 Description

The Vitamin Deficiency Detection System is an intelligent healthcare support application designed to identify possible vitamin deficiencies based on user symptoms. The system allows users to enter symptoms through a web interface, and the application analyzes the symptoms using machine learning techniques to predict potential vitamin deficiencies.

The system helps users become aware of possible nutritional deficiencies and provides recommendations for improving their health through proper diet and medical consultation. The application offers a simple and user-friendly interface for symptom input and prediction results.

4.2.2 Actors and Roles

Table 4.2: Use Case — Actors and Roles

Actor	Role
User (Patient / Individual)	Enters symptoms and health-related information for analysis
Vitamin Deficiency Detection System	Processes input data and manages the overall prediction workflow
Machine Learning Model	Analyzes symptoms and predicts possible vitamin deficiencies
Dataset	Provides labeled training data used to train and evaluate the prediction model
Web Interface	Displays prediction results along with suggestions or recommendations

4.2.3 Inputs

- Symptoms entered by the user (fatigue, hair loss, weakness, etc.)
- Health information provided by the user
- Pre-trained machine learning model
- Training dataset containing vitamin deficiency symptoms

4.2.4 Process

1. The user opens the Vitamin Deficiency Detection System through the web interface.
2. The system displays a symptom input form.
3. The user selects or enters symptoms related to their health condition.
4. The system collects the symptom data entered by the user.
5. The input symptoms are preprocessed and converted into numerical features.
6. The processed data is passed to the trained machine learning model.
7. The machine learning model analyzes the symptoms and compares them with patterns from the training dataset.

8. The system predicts the possible vitamin deficiency.
9. The prediction result is generated by the system.
10. The predicted vitamin deficiency and recommendations are displayed to the user.

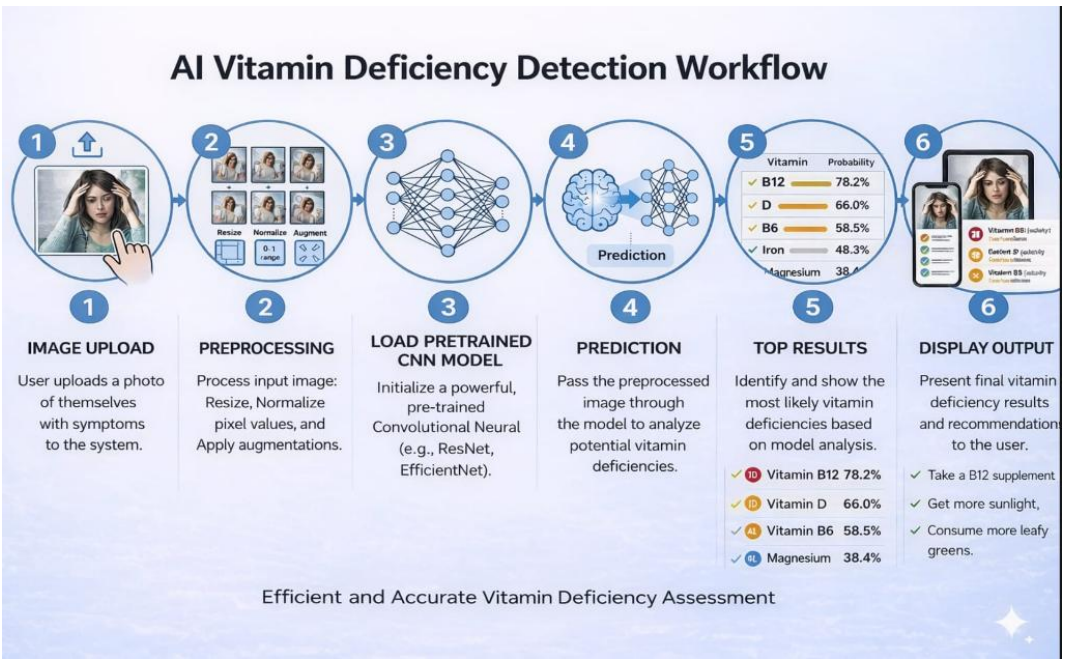

Figure 4.3: Vitamin Deficiency Detection System — Workflow Diagram

4.2.5 Real-World Example

A user experiences symptoms such as fatigue, weak bones, and frequent muscle pain. The user opens the Vitamin Deficiency Detection System and enters the symptoms through the web interface.

The system processes the symptoms and sends the data to the machine learning model. Based on the analysis, the system predicts that the user may have a Vitamin D deficiency. The system displays the result along with general health suggestions, helping the user become aware of the potential deficiency and encouraging them to consult a healthcare professional.

4.2.6 Output

• Predicted vitamin deficiency result

• Health recommendation based on symptoms

• Displayed prediction result on the web interface

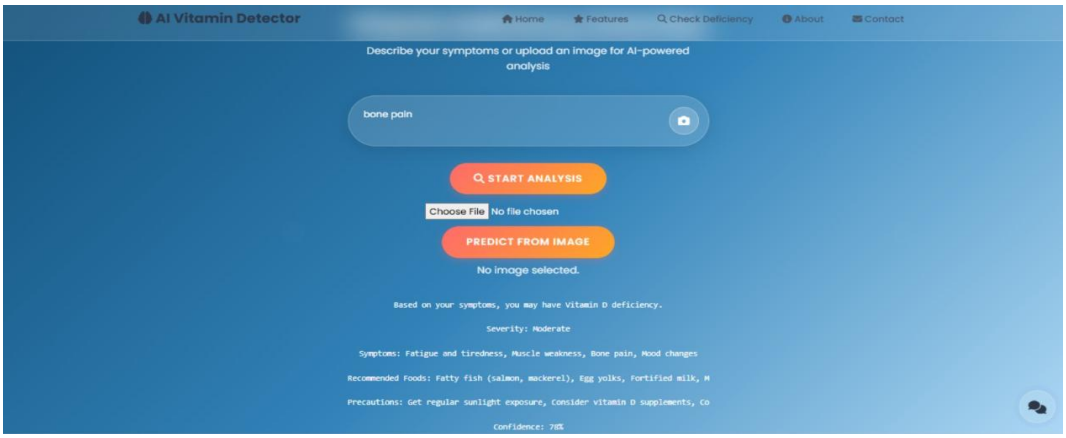

Figure 4.4: Vitamin Deficiency Detection System — Symptom Input Interface

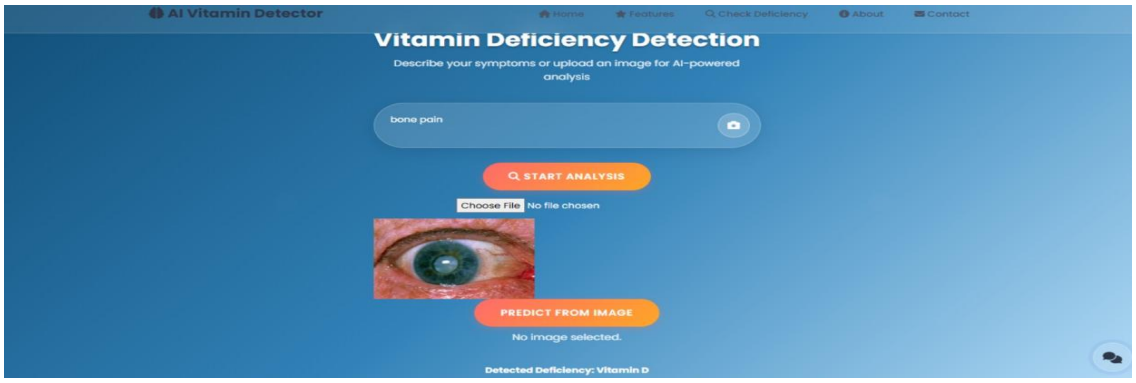

Figure 4.5: Vitamin Deficiency Detection System — Prediction Result Display

4.2.7 Advantages

• Helps users identify possible vitamin deficiencies quickly

• Provides health awareness and nutritional guidance

• Easy-to-use interface for symptom input

• Uses machine learning techniques for intelligent prediction

• Supports preventive healthcare awareness

4.2.8 Limitations

- Prediction accuracy depends on the dataset used for training
- The system cannot replace professional medical diagnosis
- Limited to symptoms included in the dataset
- Users must still consult healthcare professionals for confirmation

4.2.9 Future Enhancements

- Integration with wearable health devices for real-time health data
- Support for additional vitamin and mineral deficiency detection
- Development of a mobile application for easier accessibility
- Integration with diet recommendation systems for personalized nutrition plans

4.2.10 Conclusion

The Vitamin Deficiency Detection System demonstrates how machine learning can be applied in the healthcare domain to identify possible nutritional deficiencies based on symptoms. The system provides a simple and accessible tool for users to gain insights into their health condition.This project provided practical experience in data processing, machine learning model implementation, and web-based application development,highlighting the potential of AI-based solutions in improving healthcare awareness and early detection of health issues.

STUDENT PROFILE / RESUME

SHRILAKSHMI A S

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Objective

Aspiring AI and Software Engineer with knowledge of Python, Machine Learning, and Web Development. Developed AI-based projects such as Fabricated News Prediction System and Vitamin Deficiency Detection System. Seeking an opportunity to apply technical skills and grow in the field of Artificial Intelligence.

Education

S.No.	Qualification	Institution / Board	Year	Result
1	SSLC	State Board	2023	91.2%
2	Diploma in Computer Science & Engineering	Sri Jayachamarajendra Government Polytechnic, Bengaluru	2026 (Expected)	In Progress

Internship Experience:

AI Engineer Intern / AI Solution Developer Intern

TestHeapStack Software Solutions Pvt. Ltd., Bengaluru | December 2025 – April 2026

Projects

1. Fabricated News Prediction System

2. Vitamin Defiency Detection System

Technical Skills

Category	Skills
Programming Languages	Python (Intermediate), Java (Basic), HTML (Basic)
AI / Machine Learning	Fundamentals of Machine Learning, Natural Language Processing (NLP), Deep Learning
Databases	SQL, SQLite (Basic – Intermediate)
Libraries & Frameworks	Pandas, NumPy, Scikit-learn, TensorFlow, Seaborn, VADER, TextBlob, SentenceTransformers, FAISS, Streamlit, Plotly
Tools & Platforms	GitHub, VS Code, Jira
Other Skills	MS Office, Power BI (Basic), Windows, Linux (Basic)

Certifications

- Artificial Intelligence — Samsung Innovation Campus (SIC)
- NSS Certificate

Personal Information

Field	Details
Father's Name	Somashekar B
Date of Birth	05 April 2007
Languages Known	Kannada, English, Tamil
Gender	Female
Nationality	Indian
Marital Status	Unmarried

PHOTOGALLERY

REFERENCES

Project References (Fabricated News Prediction System)

- **Google Gemini AI** – https://gemini.google.com/app
- **ChatGPT (OpenAI)** – https://chat.openai.com
- **Flask Web Framework** – https://flask.palletsprojects.com/
- **Streamlit for Web Apps** – https://streamlit.io/
- **TensorFlow Deep Learning Framework** – https://www.tensorflow.org/
- **Keras Neural Network API** – https://keras.io/
- **NumPy Numerical Computing** – https://numpy.org/
- **Pandas Data Analysis Library** – https://pandas.pydata.org/
- **Matplotlib Data Visualization** – https://matplotlib.org/
- **Seaborn Visualization Library** – https://seaborn.pydata.org/
- **Scikit-learn Machine Learning Library** – https://scikit-learn.org/
- **Joblib (Model Saving & Loading)** – https://joblib.readthedocs.io/
- **GitHub Project Repository** – https://github.com/
- **DeepLearning.AI Deep Learning Concepts** – https://www.deeplearning.ai/

Project References (Vitamin Deficiency Detection System)

* **Google Gemini – https://gemini.google.com/app**

* **ChatGPT – https://chatgpt.com**

* **Flask Documentation – https://flask.palletsprojects.com**

* **TensorFlow Deep Learning Framework – https://www.tensorflow.org/**

* **Keras Neural Network API – https://keras.io/**

* **NumPy Numerical Computing – https://numpy.org/**

* **Pandas Data Analysis Library – https://pandas.pydata.org/**

* **Matplotlib Data Visualization – https://matplotlib.org/**

* **Scikit-learn Machine Learning Library – https://scikit-learn.org/**

* **GitHub Project Repository – https://github.com/**

* **DeepLearning.AI Deep Learning Concepts – https://www.deeplearning.ai/**

APPENDICES

Appendix A — Sample Code Snippets

A.1 Fake News Prediction — TF-IDF Vectorization and Model Training

```
# TF-IDF vectorization
tfidf_vectorizer = TfidfVectorizer(max_features=5000, stop_words='english', ngram_range=(1,2))
X_train_tfidf = tfidf_vectorizer.fit_transform(X_train_text)
X_test_tfidf = tfidf_vectorizer.transform(X_test_text)
# Logistic Regression training
logreg = LogisticRegression(random_state=42, max_iter=1000)
logreg.fit(X_train_tfidf, y_train)
y_pred = logreg.predict(X_test_tfidf)
```

A.2 Fake News Prediction — Model Evaluation

```
# Evaluate model
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
roc_auc = roc_auc_score(y_test, logreg.predict_proba(X_test_tfidf)[:, 1])

print(f"Accuracy: {accuracy:.4f}")
print(f"F1-Score: {f1:.4f}")
print(f"ROC-AUC: {roc_auc:.4f}")
```

A.3 Fake News Prediction — Saving Model and Vectorizer

```
try:
import joblib
import os
os.makedirs('models_fake', exist_ok=True)
# Save model
joblib.dump(logreg, 'models_fake/fake_news_model.pkl')
# Save TF-IDF vectorizer
joblib.dump(tfidf_vectorizer, 'models_fake/tfidf_vectorizer.pkl')
```

A.4 Fake News Prediction — Sample Predictions

```
sample_texts = X_test_text.sample(5, random_state=42)
sample_processed = sample_texts.apply(preprocess_text)
sample_tfidf = tfidf_vectorizer.transform(sample_processed)
predictions = logreg.predict(sample_tfidf)
probabilities = logreg.predict_proba(sample_tfidf)
results_df = pd.DataFrame({
'Text': sample_texts.values,
'Predicted_Label': predictions,
'Real_Probability': probabilities[:, 0],
'Fake_Probability': probabilities[:, 1]
})
print(results_df
```

Appendix B — Additional Screenshots

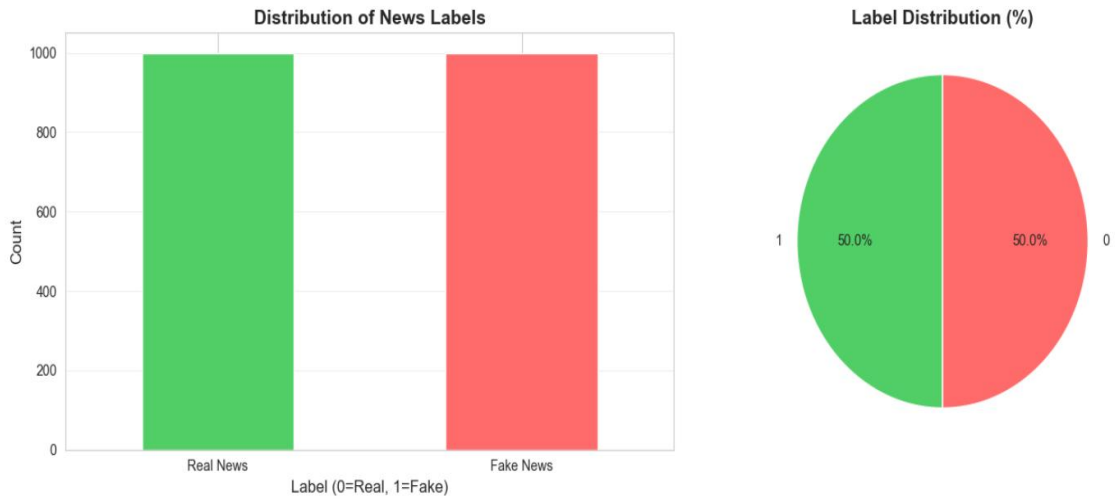

Figure B.1: Distribution of News Labels (Real vs Fake)

```
✅ Feature engineering completed!

New features: ['text_length', 'word_count', 'avg_word_length', 'uppercase_ratio', 'exclamation_count', 'question_count', 'sensational_count']

Feature statistics:
       text_length   word_count  avg_word_length  uppercase_ratio  \
count  2000.000000  2000.000000      2000.000000      2000.000000
mean    449.668000    64.742500         6.016527         0.035413
std      41.739296     9.039195         0.484490         0.012579
min     372.000000    48.000000         5.191176         0.019956
25%     428.000000    59.000000         5.565217         0.028017
50%     445.000000    64.000000         5.976077         0.031320
75%     463.000000    70.000000         6.442623         0.038924
max     569.000000    87.000000         7.041667         0.080260

       exclamation_count  question_count  sensational_count
count         2000.00000     2000.000000        2000.000000
mean             1.19900        0.105500           1.222000
std              1.24827        0.307273           1.779129
min              0.00000        0.000000           0.000000
25%              0.00000        0.000000           0.000000
50%              1.00000        0.000000           0.000000
75%              2.00000        0.000000           2.000000
max              3.00000        1.000000           7.000000
```

Figure B.2: Feature Histograms (Text Length, Word Count, Uppercase Ratio, Sensational Words)

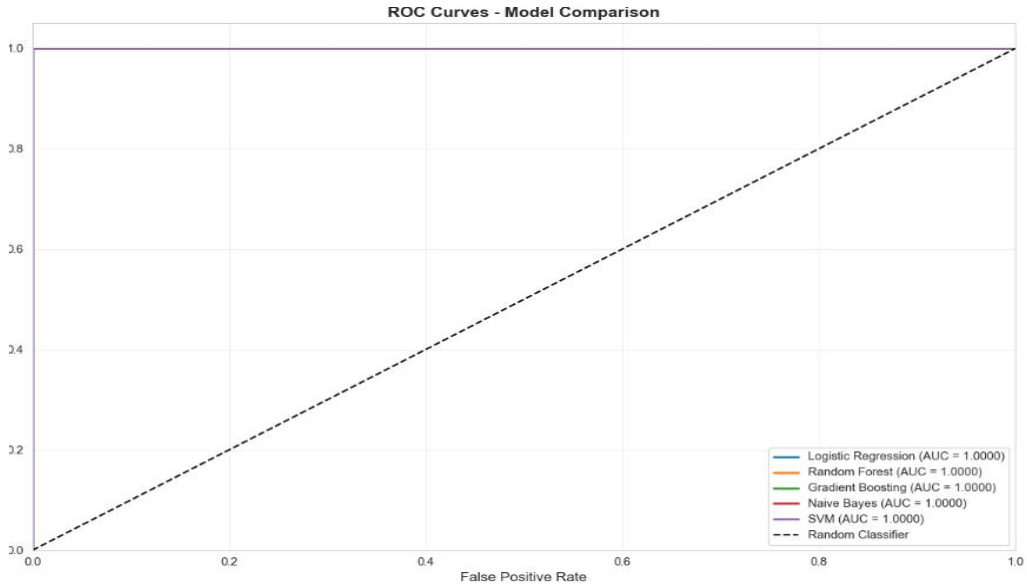

Figure B.3: ROC Curves of Different Models (Logistic Regression, Random Forest, Naive Bayes, etc.)

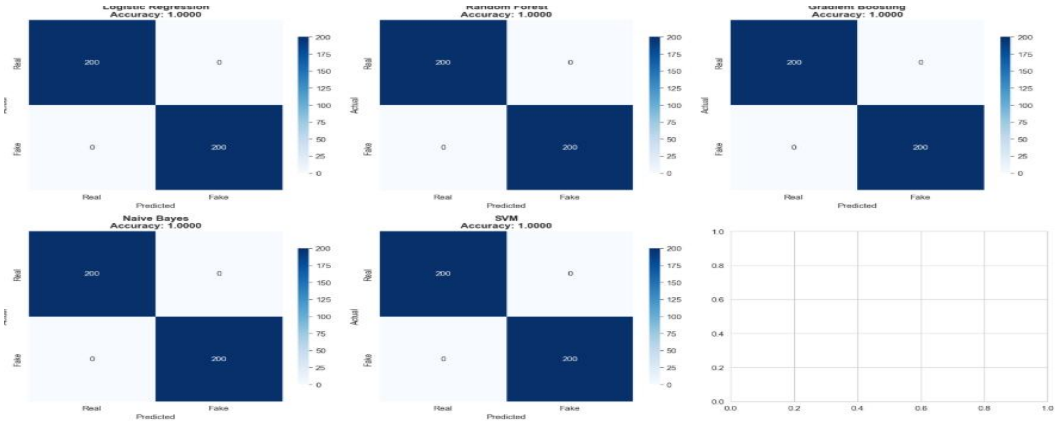

Figure B.4: Confusion Matrices of All Trained Models

Appendix C — Sample Inputs and Dataset Details

C.1 Sample Test Articles Used

Feature	Count / Details	Feature
Total Articles	~20,000+	Total Articles
Real News	~10,000	Real News
Fake News	~10,000	Fake News

C.2 Dataset Summary

Feature	Count / Details
Total Articles	~20,000+
Real News	~10,000
Fake News	~10,000
Columns	title, text, label, combined_text, processed_text